

White Paper



AWTG

5G SNAPSHOT June 2018

www.awtg.co.uk

5G Snapshot

I. Purpose of this document

This document provides the latest status update of 5G mobile standards and technology development in the UK market in Q2 2018.

II. Background

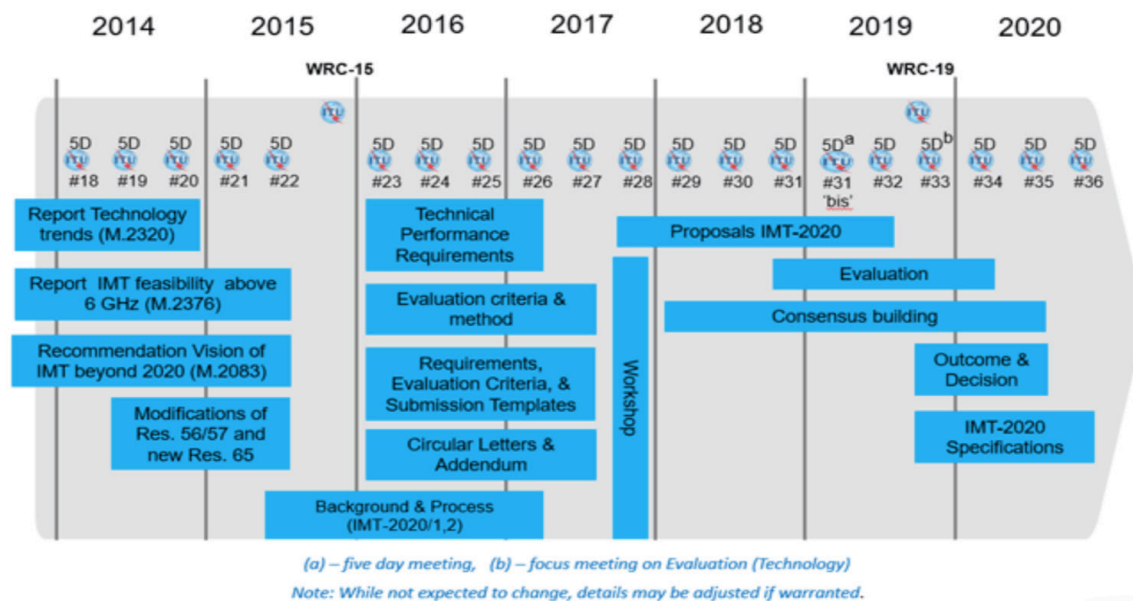
The next generation mobile technology 5G is not just the next G. It is an evolution offering high speed, ultra-low latency and reliability compared to the existing 4G version. Its development started by IMT2020 (International Mobile Telecommunication Systems as part of ITU-R => International Telecommunication Union – Radio communication Sector) in 2012. Their main vision was finalised in 2015 and aimed to develop a mobile broadband connected society. Their indicative timeline is shown on Figure 1. below.

According to IMT-2020 key performance indicators of 5G networks should deliver:

- eMBB – Enhanced Mobile Broadband (speed up to 10GBps to end users)
- URLLC – Ultra-reliable and low latency communication (sub 10ms latency, 99.999% reliability)
- mMTC – Massive machine type communication (10 times more devices per square km than in 4G)

Other collaborative research efforts such as 5G PPP (Public Private Partnership) has been established by the European Union in order to strive for innovation and foster industry research focusing on business driven KPIs.

The leading industry players such as BT, AT&T, T-Mobile, NTT Docomo, Tele2, SK Telecom, Telefonica etc. have also organised themselves into groups such as the NGMN (Next Generation Mobile Networks) Alliance with a mission is to drive standardization for the next generation mobile infrastructure, devices and service platform to meet the requirements of mobile operators. NGMN have defined the potential 5G use cases and outlined some business model opportunities behind them.



Detailed Timeline and Process for IMT-2020 in ITU-R

Figure 1. IMT-2020 timeline and process

III. The standardisation status

3GPP (3rd Generation Partnership Project) is one of the preminent standards bodies for telecommunication network technologies, including radio access, the core transport network, and service capabilities. Their working groups have been focusing on the new 5G standards specifications based on the IMT 2020 requirements and have successfully locked down the new radio interface called NR (New Radio) in June 2018.

This NR specification is part of 3GPP Release 15 and defines 5G in an NSA (Non-Standalone) environment where 5G radios will co-exist and be supported by LTE radios that will provide control plane functionality for 5G radio communications. 3GPP defines this as 5G Phase 1, where the focus is on the new radio interface to provide high data rate to end users overlaying the existing 4G radio access network and utilising its EPC (Evolved Packet Core). Release 15 specifications also includes the definition of SA (Standalone) implementation principles with 5G NR.

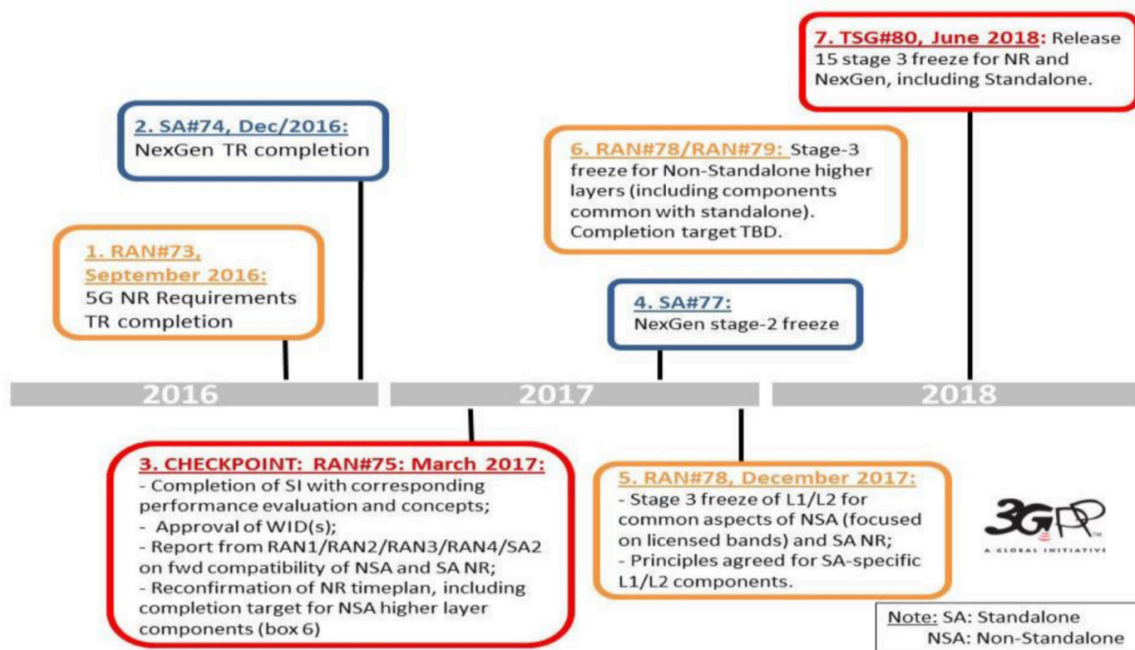


Figure 2. 3GPP 5G specification timelines for new radio interface

In Phase 2 (3GPP Release 16) 5G will include more features in SA (Standalone) enhanced implementation anchored to the new 5G core offering a full range of 5G capabilities defined by IMT2020.

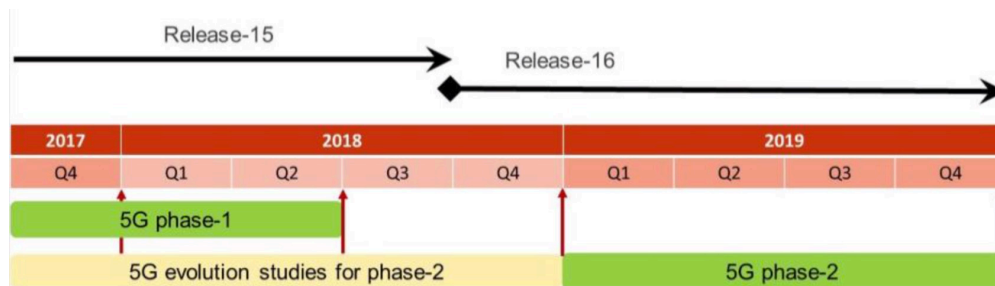


Figure 3. 3GPP Release 15 and 16 timelines

The Release 16 specifications are expected to be completed by Q4 2019 with commercial availability of equipment some 6-12 months after the specifications have been locked down.

Many vendors will have pre-standard kit available for testing purposes which is normally provided by their R&D teams, but the dependency is usually about the interconnecting interfaces development.

IV. 5G test beds and early deployment

There are several early implementations claimed as 5G deployments by service providers around the world driven by key events such as the Olympic Games and Football World Cup as well as political factors with countries and governments across the globe vying to take a lead on 5G network deployment.

Korea Telecom (KT) claimed an early implementation of 5G technology for the 2018 PyeongChang Winter Olympic games including high speed and low latency communications and showcased several use cases at the event.

Other initiatives are driving early 5G deployment in many European countries. For example, **in the UK, DCMS (Department for Culture Media and Sport) are funding the development of various 5G test beds across the country.** These test beds are focusing on different regions and exploring a variety of use cases that 5G will deliver.

The UK government's 5G strategy strives to challenge businesses, local authorities and communities to address some of the economical and practical challenges of the 5G deployment.

Thus far six test beds have received DCMS grant funding for building their solutions:

1. 5G Rural First: Led by Cisco, and Strathclyde University in Scotland, focusing on developing a test bed for rural communities based on agriculture, broadcasting and utilities. The test bed is based on Orkney Islands in Scotland addressing some of the challenges for smart farming.

2. 5G Smart Tourism: Led by WECA (West of England Combined Authority) focusing on Augmented Reality (AR) and Virtual Reality (VR) technology in major tourist attractions in historical Bristol and Bath city.

3. Worcestershire 5G Consortium: Led by WLEP (Worcestershire Local Enterprise Partnership) focusing on Industry 4.0 use cases, specifically looking at preventative and assisted maintenance, using robotics and big data analytics using AR technology.

4. Liverpool 5G test bed: Led by Sensor City focusing on health suppliers such as NHS, academia researchers and a key leading vendor. The main use case this consortium will focus on is IoT in healthcare and will address challenges on how to reduce the digital divide, patient monitoring, remote doctors etc.

5. AutoAir: Led by Airspan, focusing on CAV (Connected and Autonomous Vehicles) based in Millbrook. Will also look at 5G network transportability for road and rail transport systems as well as small cells in the sub 6GHz spectrum.

6. 5G Rural Integrated Test Bed: Led by Quickline communications, focusing on rural applications in agriculture, tourism and connecting poorly served communities using shared spectrum in Cumbria, Northumberland, North Yorkshire, Inverness-shire, Perthshire and Monmouthshire areas.

AWTG is a member of Worcestershire 5G Consortium and responsible for developing network design and delivering the physical implementation of the test bed in collaboration with other key members such as Huawei and 5GIC. AWTG will also run and operate the test bed throughout the duration of the grant fund period and support the development of the key use cases under test. Furthermore AWTG is also involved and supporting other 5G test beds across the UK both from a design and deployment perspective as well advising on how to support their 5G Use case testing regimes.

V. Deployment Challenges

Mobile operators face a number of hurdles and challenges that need to be overcome before they can start to deploy 5G networks in any meaningful scale.

- Availability of equipment and end user devices.
Whilst a lot of ground work has already been completed by equipment manufacturers, they will not be able to complete 5G equipment development until standards are agreed and formally published. 3GPP Release 15 (Phase 1) has only just been locked down in June 2018 meaning of equipment vendors are expected to start early delivery of products in Q2 2019. Timescales have been expedited thanks to many manufacturers having completed their hardware platforms in advance of the standards, with only software and firmware components needing to be completed once the standards are locked.
- Core network will require a major update (the current implementation is based on 4G standards), and 5G Core will be in a fully virtualised environment, which requires powerful virtualisation platforms.
- End user devices, including smartphones are not expected before Q4 2019, however early devices such as modems are going to be available sooner.
- In terms of implementation there are several challenges such as the fibre backhaul network needs a major update in terms of speed and capacity to support gigabit speed to each end user device, site permission for installation for outdoor sites cost and time implications just to mention a few.

In addition to the practical challenges listed above, it is still not clear how the business case for 5G deployment will be resolved as operators face uncertain costs with no clear path as yet for return on investment. Most parties are agreed that deployment costs are likely to be significantly above those of 4G once licence costs are stripped out, as 5G will require a significant increase in site density if it's to deliver on the projected capacity increase and coverage needed for the key 5G use cases such as Autonomous Cars, eMBB and Massive IoT.

Beyond the initial deployments in key urban centres such as London and Edinburgh, operators are likely to slowly phase in 5G across the remainder of the country as the demand and business case for those areas builds and justifies the investment AWTG understands that a clear business case is needed to enable wide scale 5G deployment and have been working to develop a cost model to enable operators to understand how costs would be impacted through a number of different deployment scenarios. Anyone wishing to discuss and review the findings of our current modelling should contact **Rohan Martin** at rohan.martin@awtg.co.uk.

VI. End user devices status

The major network equipment vendors also manufacture and account for the vast majority of end user devices and customer premise equipment. As such the end user devices development usually happens in parallel to network equipment development as both have the same dependencies in terms of standardisation and the availability of chipsets.

Early proof of concepts and prototype devices are available for use by operator R&D groups and some units have been provided to the test beds and other research efforts previously highlighted. However, the first commercial 5G capable mobile device is not expected to be available before Q2 2019 and possibly later than this subject to progress in terms of device development and mass manufacturing dependencies.

VII. AWTG's 5G experience

Since 2014, AWTG has been involved in several 5G development activities and now enjoys an active role in several major UK test bed facilities such as the University of Surrey 5G Innovation Centre (5GIC) test bed and the Worcestershire 5G Test Bed Consortium as a system integrator.

At 5GIC, AWTG have helped to deliver one of the most extensive and comprehensive 5G test facility in Europe, capable of testing all aspects of the new 5G network architecture across a ultra-dense network of sites, delivering a range of deployment scenarios from greenfield masts through to urban rooftops and wall mounted installations. As an associate member of 5GIC, we continue to actively work with the 5GIC team to develop commercial grade 5G network systems and features that will enable the delivery of 5G use cases in different industries such as Health, Internet of Things, Emergency, Agriculture, Automotive etc.

AWTG is also working with Kings College (KCL) London on several different 5G use cases focusing on AR (Augmented Reality) and VR (Virtual Reality) areas.

VIII. Conclusion

Whilst 5G technology is evolving to become more mature technology and mobile operators are facing a number of deployment challenges. Some of these technology and commercial challenges are outlined in the document. 5G has the potential to revolutionise our lives, the way we interact with one another and we engage with increasingly sophisticated and interconnected devices. Even though the technology is emerging, it will transform the vertical industries such as automotive, agriculture, health, transport and manufacturing just to mention a few.

Early 5G test beds implementations across the UK, will showcase the capabilities of 5G network technologies and demonstrate the benefits that new use cases can deliver.

IX. About AWTG

AWTG is an experience provider of 5G networking and IoT services and solutions. We have a range of service propositions designed to meet all your networking and IoT requirements, with a bespoke 5G service proposition for those clients looking to establish strong network foundations and capability to take full of advantages of future 5G enabled services.

ENGINEERING SERVICES

CONSULTANCY SUPPORT

- 5G Business Case
- 5G Core and RAN Expertise
- 5G Strategy
- Testbed design and deployment
- 5G Use case development

PLANNING AND DESIGN

- Spectrum Utilisation Measurements
- CW testing
- Indoor and Outdoor Coverage Predictions
- CAD site construction drawings

INSTALLATION AND COMMISSIONING

- Cabling
- Civil Works
- Power Works
- Fibre Works
- Network Acceptance
- User Acceptance Tests

PERFORMANCE OPTIMISATION

- Network monitoring
- Technical support
- Trouble ticketing
- Fault resolution
- Performance reporting

For more information please visit www.awtg.co.uk or contact enquiries@awtg.co.uk